

Question 1

Correct Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats. Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

```
1  #include <stdio.h>
2  int main() {
3      int n;
4      scanf("%d", &n);
5      int a[n];
6      for (int i = 0; i < n; i++) {
7          scanf("%d", &a[i]);
8      }
9      for (int i = 0; i < n; i++) {
10         for (int j = i + 1; j < n; j++) {
11
12             if (a[i] == a[j]) {
13                 printf("%d", a[i]);
14                 return 0;
15             }
16         }
17     }
18     return 0;
19 }
20
```

	Input	Expected	Got	
✓	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✓
✓	5 1 2 3 4 4	4	4	✓
✓	5 1 1 2 3 4	1	1	✓

Passed all tests!

Correct

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11             if (a[i] == a[j]) {
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Question 1 Correct Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second

array Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```
1 #include <stdio.h>
2 int main() {
3     int T;
4     scanf("%d", &T);
5     while (T--) {
6         int n1, n2;
7         scanf("%d", &n1);
8         int a[n1];
9         for (int i = 0; i < n1; i++) {
10             scanf("%d", &a[i]);
11         }
12         scanf("%d", &n2);
13         int b[n2];
14         for (int i = 0; i < n2; i++) {
15             scanf("%d", &b[i]);
16         }
17         int i = 0, j = 0;
18         int first = 1;
19         while (i < n1 && j < n2) {
20             if (a[i] == b[j]) {
21                 if (first == 0) {
22                     printf(" ");
23                 }
24                 printf("%d", a[i]);
25                 first = 0;
26                 i++;
27                 j++;
28             }
```

```

29 |         else if (a[i] < b[j]) {
30 |             i++;
31 |         }
32 |         else {
33 |             j++;
34 |         }
35 |     }
36 |     printf("\n");
37 | }
38 |
39 | return 0;
40 | }
41 |

```

	Input	Expected	Got	
✓	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✓
✓	1 6 1 2 3 4 5 2 1 6	1 6	1 6	✓

Passed all tests! ✓

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Example

Input:

```
1
3 10 17 57
6 2 7 10 15 57 246
```

Output:

```
10 57
```

Input:

```
1
6 1 2 3 4 5 6
2 1 6
```

Output:

```
1 6
```

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

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```
1 #include <stdio.h>
2
3 int main() {
4     int T;
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11         int a[n1];
12         for (int i = 0; i < n1; i++)
13             scanf("%d", &a[i]);
14
15         scanf("%d", &n2);
16
17         int b[n2];
18         for (int i = 0; i < n2; i++)
19             scanf("%d", &b[i]);
20
21         int i = 0, j = 0;
22
23         while (i < n1 && j < n2) {
24             if (a[i] == b[j]) {
25                 printf("%d ", a[i]);
26                 i++;
27                 j++;
28             }
29             else if (a[i] < b[j])
30                 i++;
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31         else
32             j++;
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35     printf("\n");
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38     return 0;
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	Input	Expected	Got	
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✓	1 6 1 2 3 4 5 2 1 6	1 6	1 6	✓

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Question 1 | Correct | Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that $A[j] - A[i] = k$, $i \neq j$.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as $5 - 1 = 4$

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int main() {
4      int n, k;
5      scanf("%d", &n);
6
7      int a[n];
8
9      for (int i = 0; i < n; i++)
10         scanf("%d", &a[i]);
11
12     scanf("%d", &k);
13
14     int i = 0, j = 1;
15
16     while (i < n && j < n) {
17         if (a[j] - a[i] == k) {
18             printf("1");
19             return 0;
20         }
21
22         if (a[j] - a[i] < k)
23             j++;
24         else
25             i++;
26
27         if (i == j)
28             j++;
29     }
30
31     printf("0");
32
33     return 0;
34 }
35

```

	Input	Expected	Got	
✓	3 1 3 5 4	1	1	✓
✓	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✓
✓	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✓
✓	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✓

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